

Breath Tools: Exploring the Effects on Adherence and User Experience of 4 Sounds Assisting Runners with Coupling Breath to Steps

Vincent van Rheden
vincent.vanrheden@plus.ac.at
Human Computer Interaction
Division, Department of Artificial
Intelligence and Human Interfaces,
University of Salzburg
Salzburg, Austria

Eric Harbour
Thomas Finkenzeller
eric.harbour@plus.ac.at
thomas.finkenzeller@plus.ac.at
Department of Sport and Exercise
Science, University of Salzburg
Hallein/Rif, Austria

Alexander Meschtscherjakov
alexander.meschtscherjakov@plus.ac.at
Human Computer Interaction
Division, Department of Artificial
Intelligence and Human Interfaces,
University of Salzburg
Salzburg, Austria

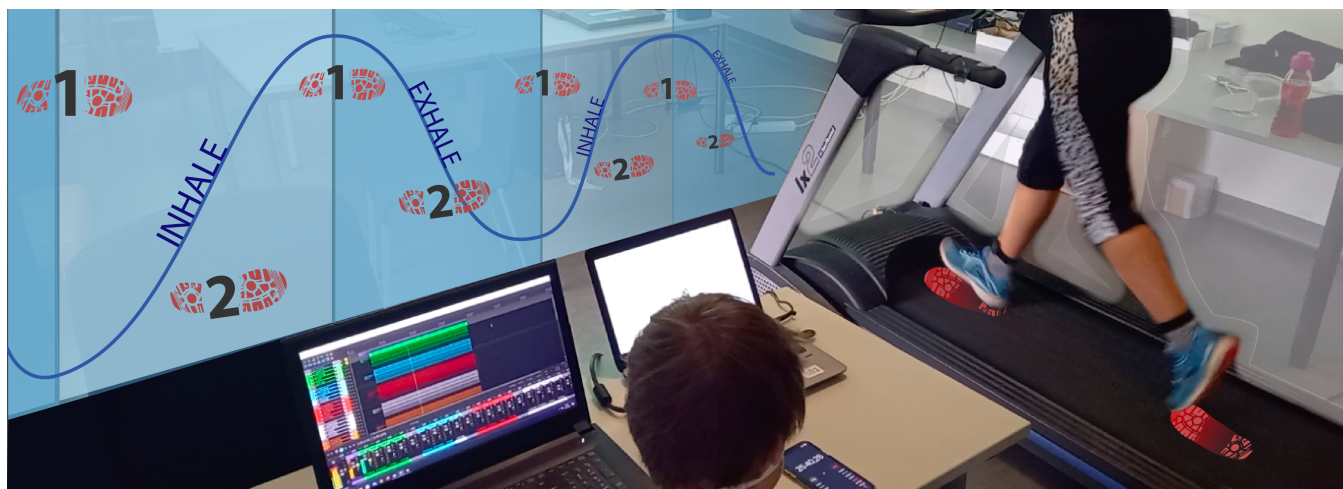


Figure 1: Evaluating various sounds to couple breath to steps while running; top left illustrating Locomotor-respiratory coupling (LRC) in a 2:2 ratio; two steps for inhalation, two steps for exhalation.

ABSTRACT

Running is a popular sport worldwide. Locomotor-Respiratory Coupling (LRC), a breathing technique in which the breath is coupled to steps, can positively impact running experience and economy. LRC may be difficult to learn and maintain during a run. Sound cues are a promising method to guide LRC while running. We designed four guiding sounds and evaluated these in a preliminary study ($N=9$). All sounds supported participants with LRC (on average $74.9 \pm 16.2\%$ of time attached to the LRC rhythm). Based on our findings we advocate sound guidance for LRC and present directions to improve adherence and run experience.

CCS CONCEPTS

• **Human-centered computing** → *Sound-based input / output; Empirical studies in HCI.*

KEYWORDS

Auditory interface, sports, running, breathing, coupling, human-computer interaction, user experience

ACM Reference Format:

Vincent van Rheden, Eric Harbour, Thomas Finkenzeller, and Alexander Meschtscherjakov. 2023. Breath Tools: Exploring the Effects on Adherence and User Experience of 4 Sounds Assisting Runners with Coupling Breath to Steps. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585736>

1 INTRODUCTION

Running is one of the most popular forms of physical activity worldwide. In times of Covid-19 it even grew in popularity [47]. It is not surprising as running is an easily accessible form of exercise, requiring minimal equipment and (social) organization, and contact with others [23]. HCI Researchers have explored various ways to

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585736>

support the runners (e.g. [24, 34, 35, 42]) and running technique (e.g. [14, 18]).

Breath is – apart from striding – an important aspect constituting the experience of running. It is a determining factor on the feeling of a run ‘going well’ [19]. Respiration is not only an important indicator for perceived workload [36], but may also be consciously controlled in order to enhance the running experience [17].

Breathing rate (BR) synchronized to step rate (SR) is called *Locomotor-Respiratory Coupling* (LRC), also referred to as rhythmic breathing or coupled breathing [8, 17]. This typically is done in whole-integer ratios, counted in steps per inhalation and exhalation. For example, one could use a 2:2 ratio, implying that the runner performs two steps per inhalation and two steps per exhalation, as illustrated in figure 1.

LRC has been shown to have a variety of psycho-physiological benefits, such as decreased oxygen consumption, increases running economy, and reduction of dyspnea (i.e. breathlessness) [4, 11, 21, 49]. Especially female runners could benefit from LRC, since they are especially susceptible to respiratory limitations and exertional dyspnea, or breathlessness, which refers to the sensation of uncomfortable breathing [1]. Therefore we focus in this preliminary study on female runners.

There is limited literature available on teaching and learning LRC. Coates and Kowalchik [8] provide an extended step-wise protocol to learn LRC. They recommend that runners (1) breathe while laying on the back, practicing the breathing rhythm by counting, (2) tap with the feet to the count and mimic the step/breath rhythm, (3) practice the rhythm while walking, and (4) apply it while running. They caution runners to not proceed to the next step until they are confident in the preceding one. We argue that a well-designed guiding technology during running itself could support especially novice runners in this lengthy and time-consuming process. This would shift learning from pre-run to during run to increase effectiveness and enjoyment of utilizing the breathing technique.

In this study we chose to explore sound as the guidance modality for two main reasons; first, sound might be safer than visual guidance [13], since vision is occupied with observing the environment. Moreover, tactile interfaces have reported low adherence [50], perhaps being hard to perceive in a dynamic setting such as running (vibrations from step impact, sensations of wind or rain on the skin). Second, sound is highly suitable to portray stable rhythms (e.g. metronomes in music) and people show an innate tendency to follow auditory rhythms with bodily movement [7, 10]. In this paper we explore and evaluate LRC guiding sounds in a user study (N=9), aimed to compare four distinct LRC guidance sounds with respect to adherence and User Experience (UX). Overall we address the following research questions:

RQ1: Can we support LRC through sound during the run?

RQ2: What is the effect of LRC guiding sounds on the runners User Experience with this system?

The contributions of this paper are twofold; first, we report the results of the user study concluding that sound guidance is suitable to support LRC during running. Second, we provide insights on the design of the sounds as well as on their effect on UX.

2 RELATED WORK

Humans naturally tend to synchronize their bodily motion with acoustic rhythms [7, 10]. As such it is not surprising that in sports the utilisation of sounds and sonification has been in the focus of many studies as shown by these literature reviews [25, 43, 52]. Especially from a SportsHCI perspective the utilisation of sound as an interaction modality is promising since it does not interfere with eyesight, which is crucial for the safety of an athlete, as argued by [13].

Next to movement sonification (as demonstrated in e.g. [12, 46, 55]), sound can also be used purely instructively without feedback to entrain repetitive movement tempo [20, 29, 37]). Others have researched the many ways that external sound or music can influence physical activity itself [48, 51].

2.1 Systems supporting Running Rhythms

2.1.1 Stride Rhythm. There has been substantial work on the effects of sound and music on running in sport science [37, 38] and HCI [39] using a metronome to affect SR and or its variability. Moens and colleagues [33] developed DJogger, an application aimed to maintaining step-music synchronization. Kitahara et al. [27] successfully demonstrated how to support running at a more stable pace by synchronizing music playback speed with runner’s pace. A commercial example is Counterpace¹, a mobile application aiming to support the synchronization with steps and heart beats using audio cues. Spotify offers multitude of playlists containing songs selected based on BPM to match running SR.

2.1.2 Breathing Rhythm. Similar to other bodily movements, humans have a tendency to follow auditory rhythms with their breathing [15]. Auditory instruction of breathing has been utilized in various areas within HCI, such as meditation applications and breathing games [28, 41, 44, 45]. In these applications sound is typically a secondary feedback modality, redundant to the visual modality. While the aforementioned work, as well as existing smartphone apps (i.e., Awesome Breathing² and Breathwrk³) provide paced breathing instruction through sound, they lack the ability to instruct breathing *based on* and in combination with step rate, which is an essential aspect of LRC. The sounds and cues provided by these applications are inspiring, yet are specifically designed for meditation and calming purposes and aesthetically don’t fit the energetic and dynamic context of running.

In the context of running, van Rheden et al. evaluated sounds to instruct paced (constant) BR, reporting that various levels of information richness of sound led to good adherence. However, sounds that instructed both inhalation and exhalation that faded in and out was preferred by participants [53]. In the current work we expand on this by exploring the adherence to LRC sounds (adding step sounds) and instruction adherence to prescribed BR synchronized by SR (ratio coupling).

2.1.3 Systems supporting LRC. A variety of studies have utilized sonic breathing instruction during cycling [2, 20, 22]. We highlight

¹Counterpace, Pulson, Inc., California, USA, <https://counterpace.com/>

²Awesome Breathing: Pacer for M, Awesome Labs LLC, New York, USA, <https://apps.apple.com/us/app/awesome-breathing-pacer-timer/id1453087953>

³breathwrk, Breathwrk Inc., Los Angeles, USA, breathwrk.com

[21] who utilized metronome sounds to guide BR or cadence and examine the effects of each on LRC occurrence. They found that both forms of sound guidance (pacing BR or cadence) entrained the other (e.g., paced BR entrained pedaling cadence), leading to higher spontaneous LRC. This emphasizes the inherent tendency of humans to synchronize to external audio cues, especially during rhythmic exercise.

Valsted et al. [50] designed a mobile system called Strive to assist runners with LRC while running using haptic instructions. They explored providing vibration patterns based on step count to instruct either inhalation or exhalation (i.e. vibration on each step during the inhalation, or vibration on each step during the exhalation). Differently from [53] they did not evaluate instruction on both inhalation and exhalation due to the low resolution of the tactile technology they used. They found success rate of around 25% for their system. Our work is distinguished from Strive by using the auditory modality for breath guidance, and instructing both inhalation and exhalation as this was shown to be subjectively more effective for participants when guiding the breath through sound [53].

3 STUDY: COMPARING LRC GUIDANCE SOUNDS IN THE LAB

Our user study focused on comparing four LRC guidance sounds with respect to adherence and UX in the lab⁴. First we present the design and rationale of the LRC guidance sounds. Then we describe the study setup covering procedure, instruments, data processing, and finally study results.

3.1 Four LRC Guidance Sounds

We designed four different LRC guiding were designed using REAPER⁵ sound design software. The first sound (*BreathFlow*) consisted of actual breathing recordings to indicate inspiration and expiration. This sound was intended to be maximally similar to real breath to maximize information richness of this sound, which is shown to be advantageous for UX [53]. Steps were paced by tambourine beats. The second sound — labelled *PingPong* — was most simplistic and contained metronome-like high-pitched 'pings' and lower-pitched 'pongs', similar to the Full Metronome sound presented by [53]. *PingPong* was designed to explore hybrid LRC guidance, which combines breath and step indicators in a single sound (thus no tambourine beats are needed). It does so by alternating the higher pitched 'pings' when taking steps during inhalation and with 'pongs' indicating steps during exhalation. This sound deliberately had no continuous sound stretched over the breath. People often listen to music during their runs. As such, the final two sounds were designed to include additional musicality and potentially service different music preferences. The third sound (*RockJam*) was a rock-inspired sound. It consisted of two electric guitar strikes: the first strike having a higher tonality to indicate inspiration, and the second a lower tonality to indicate expiration. The final sound was a classical music-inspired sound labeled *Strings* consisting of two string arpeggios with similar two-tone indications. Both *RockJam*

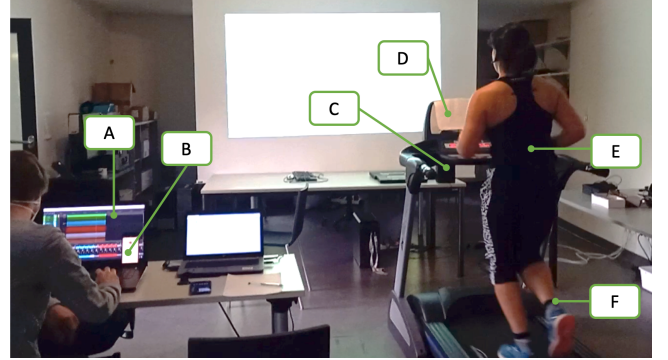


Figure 2: Instruments: (A) laptop with REAPER sound software, (B) phone with Movesense application, (C) blue-tooth speaker, (D) treadmill, (E) Hexoskin sensor shirt, (F) Movesense IMU

and *Strings* sounds indicated steps with tambourine beats identical to *BreathFlow*.

3.2 Participants

We recruited a convenience sample of nine female runners. Selection criteria were as follows:

- participants do not have prior experience with LRC during running
- participants should self-identify themselves as beginner to intermediate runners
- participants are able to run for at least 20 minutes without a break

Participants were 18-30 years old (24.3 ± 3.3) and reported running an average of 1-2 sessions per month. The average running distance was 5 km (4-7.5km) with an average running time of 35min (25-50min). None of the participants indicated prior experience with LRC. We acknowledge the number of participants and the primary focus on female runners as a limitation for the generalizability of the findings for this preliminary study, yet we are happy to support this understudied subgroup [9].

3.3 Study Procedure

In this within-subjects design study, participants ran on an indoor treadmill with each LRC instruction sound for five minutes. Figure 2 visualises the experimental setup and instruments (outlined in the following subsection 3.4).

First, participants answered a pre-participation questionnaire including basic demographics, running and sport behaviour, and breathing experience. Then, they were informed about the exact study procedure. Next, they were guided through a seated sound familiarization protocol, in which the sounds were played and participants practiced to follow along with their breath and tapping their feet. All participants successfully showed to be able to synchronize with all sounds.

⁴For this research ethical approval was granted by the Ethics Committee of the University of Salzburg.

⁵REAPER v6.28, Cockos Incorporated, Rosendale, NY, USA

After the familiarization, participants were fitted with a Hexoskin⁶ smart shirt according to manufacturer's specifications and performed a 10 min warm-up on the treadmill. During the first five minutes they were encouraged to select a comfortable speed; this speed was maintained during the rest of the experiment. A *talk test* [54] was performed during the second half of the warm-up to confirm the comfort of the self-selected speed. The mean BR and SR of the participant was derived from the last minute of the warm-up as a representation of each individual's natural running rhythm at physiological steady state. This measurement was used to individualize the instructed LRC ratio (step rate/breathing rate) of each sound. Finally, participants were asked to indicate their perceived rate of exertion from 6 to 20 on the BORG scale [6] and after their response the treadmill was stopped to have a three-minute break before starting the experiment.

Participants ran for five minutes with each LRC sound, which were introduced in order of information-richness to maximize initial understanding. During the first minute no sound was played to allow participants to achieve some comfort with the treadmill speed and their personal rhythm. Then, the LRC instruction sound was played for four minutes. Hereafter the sounds were muted and participants rated their perceived exertion whilst running. We aimed to monitor and limit fatigue as it might affect adherence to breathing instructions. Thus, we established a threshold value of 16 to avoid overexertion and maintain comfortable physiological steady-state. None of the participants reached this threshold during the experiments.

After each of the sound conditions, participants rested for three minutes to recover and minimize the effect of fatigue. Hexoskin data were used to confirm that participants' heart rate lowered to 120 beats per minute before each next sound condition. During this resting phase they completed an adapted meCUE questionnaire (see 3.4.2) [32] to evaluate UX. A semi-structured interview was conducted after the runs, which included questions about the perceived usefulness, usability, and UX with the different sounds.

3.4 Setup and Instruments

Figure 2 provides an overview of the study setup. While participants ran on the treadmill breathing instruction sounds were produced in REAPER and played via Bluetooth speakers facing the treadmill. Sounds for all potential ratios were prepared in advanced. Thus, the researcher only needed to enter the respective sound including the individualised SR and LRC ratio into REAPER and replay the sound for the participant. During the run participants were wearing the Hexoskin smart shirt and Movesense inertial measurement unit.

3.4.1 Hexoskin Smart Shirt. To answer RQ1, we used metric of *adherence*, following [53]. Adherence is an objective measure to quantify *how well* participants can execute the LRC ratio instructed by the guidance sounds, and as such how effectively the sounds supported LRC. The Hexoskin smart shirt was used to gather raw breathing rate and movement data from which breathing rate, step rate, and LRC adherence was calculated (see details below). Hexoskin Smart Shirt (Hexoskin) was used to gather respiration (dual

thoracic and abdominal stretch sensors, 2 channel respiratory inductance plethysmography, 16-bit, 128 Hz), heart rate (1 Ch ECG, 12bit) and accelerometry (3-axis, 64Hz, range $\pm 16g$) data. Captured respiration data was processed using a custom-built algorithm based on [16] in MATLAB⁷. Step detection from Hexoskin accelerometer data was performed using an adapted version of the algorithm by Benson et. al [3]. LRC ratio was calculated on a per-breath basis using the quotient of average SR (over one breath cycle) and instantaneous BR. Sound instruction adherence was calculated similar to [53]. Attachments and detachments were defined a priori as five or more consecutive breaths inside or outside, respectively, of five percent of the instructed LRC ratio. For example, for an instructed LRC ratio of five, an attachment was flagged if five consecutive breaths had a ratio between 4.75-5.25 steps per breath. These were calculated from exact sound start to sound end for all conditions. We chose to report the percentage of trial time *attached* or *detached* (versus total trial time). Note that participants also can be in a state in which they were neither attached or detached from the guidance, accounting for the remaining percentage of time.

3.4.2 User Experience Questionnaire. To answer RQ2 we capture the UX of running with the four LRC guidance sounds, we utilized an adapted version of the meCUE questionnaire [32]. We adopted the approach outlined in [53] to adapt the meCUE questionnaire according to the meCUE handbook to fit the context of the research. Shortening was essential for participants to be able to timely answer questions in between each run. The scale contained six items that represent *Effectiveness*, *Efficiency*, *Intention to Use*, *Negative Emotion*, *Positive Emotion*, and *Aesthetics*.

- The breath and steps can be easily adjusted to this sound specification (*Effectiveness*)
- I consider the use of this sound during running to be absolutely useful (*Efficiency*)
- If I could, I would use this sound every day while running (*Intention to Use*)
- Running to this sound annoys me (*Negative Emotion*)
- Running to this sound exhilarates me (*Positive Emotion*)
- I find this sound attractively designed (*Aesthetics*)

In order to have an high degree of measurement precision, the items were answered on a 11 point scale, ranging from 0 (totally disagree) to 10 (totally agree) with 5.5 being neutral.

4 RESULTS & DISCUSSION

In this section we discuss the findings of this study, and suggest improvements on the sounds and directions for future systems implementing the sounds, as well as suggestions for followup studies.

Ratings of perceived exertion did not exceed 16/20 over the whole study. The average LRC ratio was 5.2 ± 1.2 steps per breath, and heart rate 159.01 ± 12.10 bpm. Table 1 shows the adherence data and collected UX data for each sound condition.

⁶Carre Technologies, CAN, <https://www.hexoskin.com/>

⁷MATLAB Version 2022a, MathWorks, Inc., Natick, USA)

Sound	Adherence		User Experience					
	%attached	%detached	Effectiveness	Efficiency	Intention to Use	Negative Emotion	Positive Emotion	Aesthetics
BreathFlow	77.3±10.9	2.9±1.6	8.1±2.3	8.1±1.7	4.6±2.7	2.2±1.7	5.2±2.6	6.2±2.6
PingPong	74.6±12.8	6.3±3.6	8.2±2.4	8.2±1.5	4.7±2.4	2.1±1.7	5.4±2.1	6.7±2.6
RockJam	73.0±25.5	3.8±2.5	7.6±2.7	7.4±1.9	3.8±1.9	3.7±3.0	5.4±2.1	6.2±3.1
Strings	74.8±15.7	5.0±3.1	7.1±2.9	7.0±2.7	3.0±2.5	3.8±3.2	4.0±2.8	6.1±2.5

Table 1: Results per sound condition. Adherence data represented in % of time attached and detached. User Experience scores per item, ranked on a scale of 0 (totally disagree) to 10 (totally agree), with 5.5 being neutral.

4.1 Adherence

A one-way, within-subjects ANOVA with repeated measures was performed in MATLAB to assess differences in adherence (dependent variable) between sound conditions (independent variable). Significance was set to 0.05. All variables met the a priori requirements for normality, assessed via Q-Q plots and Kolmogorov-Smirnov tests, and homogeneity of variances, assessed via Mauchly's test of sphericity. Post-hoc Bonferroni-corrected pairwise comparisons were used to determine where significant differences existed between sound conditions. Overall adherence to all sounds was considered good, reflected by high percent of time attached (>70%) and low percent time detached (<8%). Almost all quickly attached to the sound guidance (3.0±3.7 s), with only two cases higher than 10 seconds. The high standard deviation (25.5) for RockJam on attachment time indicates that participants adhered less homogeneously compared to the other sounds. Participants differed moderately in attachment ($F(10, 3) = 3.37, p = 0.005$) and detachment ($F(10, 3) = 4.16, p = 0.001$). We found no meaningful differences in attachment between any of the LRC sounds. This is in line with previous van Rheden et al. [53] suggesting that sound information richness does not affect instruction adherence. Contradictory to [53], we did observe a significant difference in detachment between conditions ($F(3, 10) = 4.56, p = 0.0095$), and pairwise comparisons revealed slightly more detachment to the least information rich sound, PingPong (6.3±3.6) versus most information rich sound BreathFlow (2.9±1.6). This suggests that BreathFlow led to the best overall adherence amongst the four sounds.

4.2 User Experience

No significant difference was found amongst the sound conditions for any of the UX items. The relative high standard deviations (2.5) show the diverse individual preferences of the participants. We assign these differences partly to the individual differences of ease that participants reported in the interviews, as well as differences in music taste.

4.2.1 Effectiveness & Efficiency. The UX items *Effectiveness* and *Efficiency* scored well, ranging from 7.0±2.7 of the Strings sound to 8.2±1.5 of the PingPong sound. In terms of *effectiveness* these results are higher in comparison to results of breathing instruction sounds of van Rheden and colleagues [53]. They reported an average of 6.9 for their best scoring sound. In particular the BreathFlow (8.1±2.3) and PingPong scored higher (8.2±2.4). Interview data, revealed that, in line with [26], participants mentioned the sounds were a motivating factor: *"the noise motivated me a bit. So it wasn't that*

exhausting". For others it was a reminder to concentrate on their running rhythm.

4.2.2 Intention to Use. The *Intention to Use* was scored neutral on all sounds; the PingPong sound scored best (4.7±2.4), and Strings worst with 3.0±2.5. This item reflected *daily use* i.e. usage of the sounds for every run. From interviews we found this made participants hesitant: one participant described them as rudimentary sounds only for *"training purposes"*; for every day usage more engaging and entertaining sounds were expected. Some participants noted that it did not fit their normal running preferences like speaking to a friend (P06) or listening to a podcast (P03) while running. Nonetheless, they would be open to adapt these preferences occasionally, but not for all runs. Comparing the sounds to [53], reporting a score around 3 for paced breathing sounds, the *Intention to Use* for LRC sounds was similar for Strings and was higher for BreathFlow, PingPong, and RockJam.

4.2.3 Emotions and Aesthetics. The item on *Negative Emotion* reflected that none of the sounds were considered annoying on average. Participants reported they were least annoyed by the PingPong sound (2.1±1.7) and most annoyed by the Strings sound (3.8±3.2). This contradicts findings from Valsted et al. in which participants reported that vibro-tactile feedback was considered annoying [50]. Similarly, the Strings sound scored lowest (4.0±2.8) on *Positive Emotion*, expressed by the item probing *exhilaration*, whilst PingPong scored best (5.4±2.1). For *Aesthetics*, the PingPong sound scored highest (6.6±2.6), whereas Strings scored lowest with 6.1±2.5. Scores ranging between 4–10 show the differences in taste. Especially for the rich sounds, such as RockJam or Strings, participants highlighted that repetitiveness of the sounds can become disturbing. To improve this one could add (slight) variability in the sounds (timbre, timing, etc) to give a more 'natural' feel to the sound. Similar to *Negative Emotion*, participants ranked the *aesthetics* of the sounds about 1 point higher than the paced breathing sounds presented by van Rheden et al. [53].

4.3 LRC Guidance Design

These results of our study invite discussion on how to guide LRC, and how guidance can be enriched to support the objective effectiveness and UX.

4.3.1 Stride-Adaptive Sounds. From this study we learned that it might be difficult for participants to focus on both SR and BR instructions at the same time. A future system might employ IMU data to capture SR during the run and adapt the LRC instruction based on the current SR. This would allow runners to focus predominantly on the breathing instruction sounds. Additionally it

would allow for more variability in SR, which might be needed when moving to field settings; fatigue, inclinations, corners, etc might influence SR and increase its variability.

4.3.2 Music Integration. The RockJam sound and the Strings sound were explicitly designed to capture various music tastes, and potentially be integrated in music that supports LRC. The good adherence scores for those sounds show the potential to do so. For future studies RockJam or Strings could be integrated in music pieces that guide breathing and steps to enhance the UX. Indeed, since music is extremely popular during running to motivate and energize, weaving LRC guidance directly into the music might increase adherence and acceptance. We view the implementation of the guidance sounds into energizing and motivating music, such as displayed in DJogger [5], to have high potential to increase UX and intend to use further.

5 CONCLUSION

Based on our findings with respect to adherence to LRC we conclude that sound is a highly effective modality to guide LRC. Therefore, the auditory instruction of LRC can potentially be applied to support various types of cyclic exercise, such as cycling [40], rowing [31], and walking [30]. We have shown that providing auditory feedback for breathing instructions works in the context of sport regardless of the concrete type of sound. Our findings suggest that sounds, which are "closer" to actual breathing sounds, work slightly better in terms of adherence and do not compromise user experience in comparison to more artificial sounds. In future work we aim to explore if sounds also works in a field context. A core feature we aim to explore is sound guidance that adapts to the runner's current SR. Additionally we aim to implement the sounds into music to enhance the UX further and cater for various music preferences.

ACKNOWLEDGMENTS

We would like to thank the participants running in this study. Also we thank the students involved in this research for their efforts and valuable discussion. This research was funded by the Austrian Ministry for Transport, Innovation and Technology, the Federal Ministry for Digital and Economic Affairs, and the federal state of Salzburg under the research programme COMET – Competence Centers for Excellent Technologies – in the project Digital Motion in Sports, Fitness and Well-being (DiMo).

REFERENCES

- [1] Bruno Archiza, Michael G Leahy, Shalaya Kipp, and A William Sheel. 2021. An integrative approach to the pulmonary physiology of exercise: when does biological sex matter? *European Journal of Applied Physiology* 121, 9 (2021), 2377–2391.
- [2] Benoît G. Bardy, Charles P. Hoffmann, Bart Moens, Marc Leman, and Simone Dalla Bella. 2015. Sound-induced stabilization of breathing and moving. *Annals of the New York Academy of Sciences* 1337, 1 (2015), 94–100. <https://doi.org/10.1111/nyas.12650> arXiv:<https://nyaspubs.onlinelibrary.wiley.com/doi/pdf/10.1111/nyas.12650>
- [3] Lauren C Benson, Christian A Clermont, Ricky Watari, Tessa Exley, and Reed Ferber. 2019. Automated accelerometer-based gait event detection during multiple running conditions. *Sensors* 19, 7 (2019), 1483.
- [4] P Bernasconi, P Bürki, A Bühner, EA Koller, and J Kohl. 1995. Running training and co-ordination between breathing and running rhythms during aerobic and anaerobic conditions in humans. *European journal of applied physiology and occupational physiology* 70, 5 (1995), 387–393.
- [5] Jacob T. Bieh, Piotr D. Adamczyk, and Brian P. Bailey. 2006. DJogger: A Mobile Dynamic Music Device. In *CHI '06 Extended Abstracts on Human Factors in Computing Systems* (Montréal, Québec, Canada) (CHI EA '06). Association for Computing Machinery, New York, NY, USA, 556–561. <https://doi.org/10.1145/1125451.1125569>
- [6] Gunnar Borg. 1998. *Borg's perceived exertion and pain scales*. Human kinetics.
- [7] Winston D Byblow, Richard G Carson, and David Goodman. 1994. Expressions of asymmetries and anchoring in bimanual coordination. *Human Movement Science* 13, 1 (1994), 3–28.
- [8] Budd Coates, Claire Kowalchik, et al. 2013. *Runner's world running on air: The revolutionary way to run better by breathing smarter*. Rodale Books.
- [9] Emma S Cowley, Alyssa A Olenick, Kelly L McNulty, and Emma Z Ross. 2021. "Invisible sportswomen": the sex data gap in sport and exercise science research. *Women in Sport and Physical Activity Journal* 29, 2 (2021), 146–151.
- [10] Philip W Fink, Patrick Foo, Viktor K Jirsa, and JA Scott Kelso. 2000. Local and global stabilization of coordination by sensory information. *Experimental Brain Research* 134, 1 (2000), 9–20.
- [11] F Garlando, J Kohl, EA Koller, and P Pietsch. 1985. Effect of coupling the breathing- and cycling rhythms on oxygen uptake during bicycle ergometry. *European journal of applied physiology and occupational physiology* 54, 5 (1985), 497–501.
- [12] Andrew Godbout and Jeffrey Edwin Boyd. 2012. Rhythmic sonic feedback for speed skating by real-time movement synchronization. *International Journal of Computer Science in Sport* 11, 3 (2012), 37–51.
- [13] Andrew Godbout and Jeffrey E. Boyd. 2015. Audio Visual Synchronization of Rhythm. *Proceedings - 2015 International Conference on 3D Vision, 3DV 2015* (2015), 589–597.
- [14] Andrew Godbout, Chris Thornton, and Jeffery E Boyd. 2014. Mobile sonification for athletes: a case study in commercialization of sonification.
- [15] François Haas, S Distenfeld, and K Axen. 1986. Effects of perceived musical rhythm on respiratory pattern. *Journal of applied physiology (Bethesda, Md. : 1985)* 61 (10 1986), 1185–91. <https://doi.org/10.1152/jappl.1986.61.3.1185>
- [16] Eric Harbour, Michael Lasshofer, Matteo Genitrini, and Hermann Schwameder. 2021. Enhanced breathing pattern detection during running using wearable sensors. *Sensors* 21, 16 (2021), 5606.
- [17] Eric Harbour, Thomas Stöggel, Hermann Schwameder, and Thomas Finkenzeller. 2022. Breath Tools: A Synthesis of Evidence-Based Breathing Strategies to Enhance Human Running. *Frontiers in Physiology* (2022), 233.
- [18] Mahmoud Hassan, Florian Daiber, Frederik Wiehr, Felix Kosmalla, and Antonio Krüger. 2017. FootStriker: An EMS-Based Foot Strike Assistant for Running. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 1, 1, Article 2 (March 2017), 18 pages. <https://doi.org/10.1145/3053332>
- [19] John Hockey and Jacquelyn Allen-Collinson. 2016. *Endurance Running: A Socio-Cultural Examination*. Routledge, London, Chapter Digging in: The sociological phenomenology of 'doing endurance' in distance-running, 227–242.
- [20] Charles P Hoffmann and Benoît G Bardy. 2015. Dynamics of the locomotor-respiratory coupling at different frequencies. *Experimental brain research* 233, 5 (2015), 1551–1561.
- [21] Charles P Hoffmann, Gérald Torregrosa, and Benoît G Bardy. 2012. Sound stabilizes locomotor-respiratory coupling and reduces energy cost. *PLoS one* 7, 9 (2012), e45206.
- [22] Charles P Hoffmann, Sébastien J Villard, and Benoît G Bardy. 2011. Stabilizing the locomotor-respiratory coupling using a metronome to save energy. In *BIO Web of Conferences*, Vol. 1. EDP Sciences, 00036.
- [23] Ryan M. Hulteen, Jordan J. Smith, Philip J. Morgan, Lisa M. Barnett, Pedro C. Hallal, Kim Colyvas, and David R. Lubans. 2017. Global participation in sport and leisure-time physical activities: A systematic review and meta-analysis. *Preventive Medicine* 95 (2017), 14–25. <https://doi.org/10.1016/j.ypmed.2016.11.027>
- [24] Mark Janssen, Jos Goudsmit, Coen Lauwerijssen, Aarnout Brombacher, Carine Lallemand, and Steven Vos. 2020. How Do Runners Experience Personalization of Their Training Scheme: The Inspirin E-Coach? *Sensors* 20, 16 (2020). <https://doi.org/10.3390/s20164590>
- [25] Costas I Karageorghis and David-Lee Priest. 2012. Music in the exercise domain: a review and synthesis (Part I). *International review of sport and exercise psychology* 5, 1 (2012), 44–66.
- [26] Costas I. Karageorghis and David-Lee Priest. 2012. Music in the exercise domain: a review and synthesis (Part I). *International Review of Sport and Exercise Psychology* 5, 1 (2012), 44–66. <https://doi.org/10.1080/1750984X.2011.631026> arXiv:<https://doi.org/10.1080/1750984X.2011.631026> PMID: 22577472.
- [27] Tetsuro Kitahara, Shunsuke Hokari, and Tatsuya Nagayasu. 2015. Supporting Jogging at an Even Pace by Synchronizing Music Playback Speed with Runner's Pace. *IEICE Transactions on Information and Systems* E98.D (04 2015), 968–971. <https://doi.org/10.1587/transinf.2014EDL8199>
- [28] Jinmo Lee, Neska Elhaoui, and Rosalind Picard. 2021. AmbientBreath: Unobtrusive Just-in-Time Breathing Intervention Using Multi-Sensory Stimulation and Its Evaluation in a Car Simulator. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 5, 2, Article 71 (jun 2021), 30 pages. <https://doi.org/10.1145/3463493>
- [29] Harry BT Lim, Costas I Karageorghis, Lee M Romer, and Daniel T Bishop. 2014. Psychophysiological effects of synchronous versus asynchronous music during cycling. (2014).

- [30] Stephen H Loring, Jere Mead, and Thomas B Waggener. 1990. Determinants of breathing frequency during walking. *Respiration physiology* 82, 2 (1990), 177–188.
- [31] Donald A Mahler, Bernadette Hunter, Timothy Lentine, and Joseph Ward. 1991. Locomotor-respiratory coupling develops in novice female rowers with training. *Medicine and science in sports and exercise* 23, 12 (1991), 1362–1366.
- [32] Michael Minge and Laura Riedel. 2013. meCUE-Ein modularer fragebogen zur erfassung des nutzungserlebens. *Mensch & Computer 2013: Interaktive Vielfalt* (2013).
- [33] Bart Moens, Chris Muller, Leon Van Noorden, Marek Franěk, Bert Celie, Jan Boone, Jan Bourgois, and Marc Leman. 2014. Encouraging spontaneous synchronisation with D-Jogger, an adaptive music player that aligns movement and music. *PLoS one* 9, 12 (2014), e114234.
- [34] Florian 'Floyd' Mueller and Matthew Muirhead. 2015. Jogging with a Quadcopter. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI '15). Association for Computing Machinery, New York, NY, USA, 2023–2032. <https://doi.org/10.1145/2702123.2702472>
- [35] Florian Floyd Mueller, Frank Vetere, Martin R. Gibbs, Darren Edge, Stefan Agamolis, and Jennifer G. Sheridan. 2010. Jogging over a Distance between Europe and Australia. *UIST 2010 - 23rd ACM Symposium on User Interface Software and Technology* (2010), 189–198. <https://doi.org/10.1145/1866029.1866062>
- [36] Andrea Nicolò, Carlo Massaroni, and Louis Passfield. 2017. Respiratory Frequency during Exercise: The Neglected Physiological Measure. *Frontiers in Physiology* 8 (2017), 922. <https://doi.org/10.3389/fphys.2017.00922>
- [37] Anouk Nijs, Melvyn Roerdink, and Peter J. Beek. 2020. Cadence Modulation in Walking and Running: Pacing Steps or Strides? *Brain Sciences* 10, 5 (2020). <https://doi.org/10.3390/brainsci10050273>
- [38] Anouk Nijs, Melvyn Roerdink, and Peter J. Beek. 2021. Effects of acoustically paced cadence modulation on impact forces in running. *Gait & Posture* 90 (2021), 234–238. <https://doi.org/10.1016/j.gaitpost.2021.09.168>
- [39] Stina Nylander, Mattias Jacobsson, and Jakob Tholander. 2014. Runright: Real-Time Visual and Audio Feedback on Running. In *CHI '14 Extended Abstracts on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI EA '14). Association for Computing Machinery, New York, NY, USA, 583–586. <https://doi.org/10.1145/2559206.2574806>
- [40] David J Paterson, Graeme A Wood, Alan R Morton, and John D Henstridge. 1986. The entrainment of ventilation frequency to exercise rhythm. *European journal of applied physiology and occupational physiology* 55, 5 (1986), 530–537.
- [41] Rakesh Patibanda, Florian 'Floyd' Mueller, Matevz Leskovsek, and Jonathan Duckworth. 2017. Life Tree. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*. ACM. <https://doi.org/10.1145/3116595.3116621>
- [42] Andrzej Romanowski, Sven Mayer, Lars Lischke, Krzysztof Grudzień, Tomasz Jaworski, Izabela Perenc, Przemysław Kucharski, Mohammad Obaid, Tomasz Kosizski, and Paweł W. Wozniak. 2017. Towards Supporting Remote Cheering during Running Races with Drone Technology. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI EA '17). Association for Computing Machinery, New York, NY, USA, 2867–2874. <https://doi.org/10.1145/3027063.3053218>
- [43] Nina Schaffert, Thenille Braun Janzen, Klaus Mattes, and Michael H. Thaut. 2019. A Review on the Relationship Between Sound and Movement in Sports and Rehabilitation. *Frontiers in Psychology* 10 (2019), 244. <https://doi.org/10.3389/fpsyg.2019.00244>
- [44] Ameneh Shamekhi and Timothy Bickmore. 2018. Breathe Deep: A Breath-Sensitive Interactive Meditation Coach. In *Proceedings of the 12th EAI International Conference on Pervasive Computing Technologies for Healthcare* (New York, NY, USA) (PervasiveHealth '18). Association for Computing Machinery, New York, NY, USA, 108–117. <https://doi.org/10.1145/3240925.3240940>
- [45] Chen-Hsuan (Iris) Shih, Naofumi Tomita, Yanick X. Lukic, Álvaro Hernández Reguera, Elgar Fleisch, and Tobias Kowatsch. 2019. Breeze: Smartphone-Based Acoustic Real-Time Detection of Breathing Phases for a Gamified Biofeedback Breathing Training. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 3, 4, Article 152 (dec 2019), 30 pages. <https://doi.org/10.1145/3369835>
- [46] Jelle Stienstra, Kees Overbeeke, and Stephan Wensveen. 2011. Embodying complexity through movement sonification: case study on empowering the speed-skater. In *Proceedings of the 9th ACM SIGCHI Italian chapter international conference on computer-human interaction: facing complexity*. 39–44.
- [47] Strava. 2020. Year in Sport 2020. <https://blog.strava.com/press/yis2020/> (2020).
- [48] Frederik Styns, Leon van Noorden, Dirk Moelants, and Marc Leman. 2007. Walking on music. *Human Movement Science* 26, 5 (2007), 769–785. <https://doi.org/10.1016/j.humov.2007.07.007> Music, Movement, Sound and Time.
- [49] Nariko Takano and Hisae Deguchi. 1997. Sensation of breathlessness and respiratory oxygen cost during cycle exercise with and without conscious entrainment of the breathing rhythm. *European journal of applied physiology and occupational physiology* 76, 3 (1997), 209–213.
- [50] Frederik Mørch Valsted, Christopher V. H. Nielsen, Jacob Qvist Jensen, Tobias Sonne, and Mads Møller Jensen. 2017. Strive: Exploring Assistive Haptic Feedback on the Run. In *Proceedings of the 29th Australian Conference on Computer-Human Interaction* (Brisbane, Queensland, Australia) (OZCHI '17). Association for Computing Machinery, New York, NY, USA, 275–284. <https://doi.org/10.1145/3152771.3152801>
- [51] Edith Van Dyck, Jeska Buhmann, and Valerio Lorenzoni. 2021. Instructed versus spontaneous entrainment of running cadence to music tempo. *Annals of the New York Academy of Sciences* 1489, 1 (2021), 91.
- [52] Vincent van Rheden, Thomas Grah, and Alexander Meschtscherjakov. 2020. Sonification Approaches in Sports in the Past Decade: A Literature Review. In *Proceedings of the 15th International Conference on Audio Mostly* (Graz, Austria) (AM '20). Association for Computing Machinery, New York, NY, USA, 199–205. <https://doi.org/10.1145/3411109.3411126>
- [53] Vincent van Rheden, Eric Harbour, Thomas Finkenzeller, Lisa Anneke Burr, Alexander Meschtscherjakov, and Manfred Tscheligi. 2021. Run, Beep, Breathe: Exploring the Effects on Adherence and User Experience of 5 Breathing Instruction Sounds While Running. In *Audio Mostly 2021* (virtual/Trento, Italy) (AM '21). Association for Computing Machinery, New York, NY, USA, 16–23. <https://doi.org/10.1145/3478384.3478412>
- [54] Michaela L Woltmann, Carl Foster, John P Porcari, Clayton L Camic, Christopher Dodge, Stephanie Haible, and Richard P Mikat. 2015. Evidence that the talk test can be used to regulate exercise intensity. *The Journal of Strength & Conditioning Research* 29, 5 (2015), 1248–1254.
- [55] Jiajun Yang and Andy Hunt. 2015. Real-time sonification of biceps curl exercise using muscular activity and kinematics. Georgia Institute of Technology.